MULTIMEDIA · UNIT 04

Audio in Multimedia Systems

LIU08106 · BLISS III | RIU08106 · BRIM III

Recording · Editing · Transcription · Accessibility · Preservation

Audio is useful when it is recorded responsibly and made understandable.

What you will learn

- Explain frequency, amplitude, sampling rate, bit depth and channels.
- Calculate uncompressed PCM size; compare WAV, FLAC, MP3 and AAC.
- Record, edit and export an accessible audio resource.
- Document permissions, source/master files, transcripts and access copies.

The goal is a complete, documented audio resource.

Why audio matters in information services

- Speech conveys language, accent, tone and spoken information.
- Libraries may use audio tours, author talks and pronunciation guides.
- Archives hold oral histories, event records, broadcasts and field sound.
- Accurate descriptions and transcripts make recordings more useful.

Audio adds information that text and images alone may miss.

BLISS and BRIM: one method, different responsibilities

- BLISS example: a short library orientation narration for new students.
- BRIM example: a permitted oral-history recording described with provenance.
- Both require clear sound, access controls, searchable descriptions and transcripts.
- Archival edits must be especially transparent when context is evidence.

The technical workflow is shared; professional judgement varies.

How does sound become digital?

- A vibrating source creates changing sound pressure in air.
- A microphone converts pressure changes into an electrical signal.
- An analogue-to-digital converter measures and encodes that signal.
- The digital samples are stored and later converted for playback.

Sound → microphone → ADC → file → playback.

Frequency and pitch

- Frequency means cycles per second and is measured in hertz (Hz).
- A 440 Hz tone oscillates twice as frequently as a 220 Hz tone.
- Higher frequency generally corresponds to higher perceived pitch.
- Frequency and pitch do not tell us whether a recording is intelligible.

Frequency describes signal repetition, not recording quality.

Amplitude and loudness

- Amplitude is the size of signal variation shown in a waveform.
- Greater amplitude often sounds louder, but perception also depends on context.
- A strong waveform may contain distortion, hiss or handling noise.
- Check both the waveform and the sound through headphones.

A large waveform is not proof of good audio.

Understanding waveforms

- A waveform displays sample amplitude over time.
- Silences appear as small activity; strong speech segments vary in size.
- A flat-topped peak may be a clue that the signal clipped.
- Zoom in to identify edits, but always confirm with listening.

Use waveforms to guide listening—not replace it.

Analogue and digital recording

- Analogue tape stores changing physical magnetisation; digital PCM stores numbers.
- Digitising a tape requires suitable playback equipment and capture settings.
- Noise, playback speed and tape condition may influence the transfer.
- A digitised copy needs documentation linking it to the analogue source.

A digital conversion does not automatically preserve all original quality.

Sample rate explained

- A 48 kHz mono recording contains 48,000 samples per second.
- Common delivery/capture settings include 44.1 kHz and 48 kHz.
- Samples are measurements of a continuous signal, not complete sound waves.
- Increasing sample rate after recording cannot recreate missing original detail.

Sample rate is samples/second/channel, not the MP3 bitrate.

Aliasing and the sampling principle

- Under ideal conditions, the sample rate must exceed twice the highest signal frequency.
- Anti-aliasing filters reduce frequencies that would otherwise be misrepresented.
- For example, 48 kHz capture supports content below approximately 24 kHz in theory.
- The hardware and source determine real-world usable quality.

Capture high enough for the required signal bandwidth.

Bit depth: PCM sample precision

- 16-bit PCM represents each sample with 16 bits; 24-bit uses 24 bits.
- Greater bit depth provides more possible sample values and recording headroom.
- It is separate from compressed delivery bitrate such as 96 kb/s.
- Changing a 16-bit file into 24-bit does not restore lost capture detail.

More bits do not cure poor microphone technique.

Channels: mono versus stereo

- Mono stores one independent signal; stereo typically stores two.
- Mono is often sensible for a spoken library guide.
- Stereo may preserve useful directional context for a soundscape.
- Duplicating a mono channel into two channels does not create genuine stereo.

Select the channel arrangement based on purpose.

File-size formula for uncompressed PCM

- Approximate sample bytes = sample rate × duration × bit depth ÷ 8 × channels.
- Example: 48,000 × 60 seconds × 24 ÷ 8 × 1 = 8,640,000 bytes.
- That is about 8.24 MiB, excluding header and metadata overhead.
- Keep decimal MB and binary MiB distinct when reporting sizes.

Audio file size is predictable before compression.

Comparing stereo and mono storage

- 48 kHz, 24-bit, mono, 60 seconds: approximately 8.24 MiB.
- 48 kHz, 16-bit, stereo, 60 seconds: approximately 10.99 MiB.
- Doubling channels can outweigh a reduction in bit depth.
- Calculate the payload rather than guessing from the format label.

Duration, rate, depth and channels all affect PCM size.

Why compression matters

- Uncompressed PCM preserves the decoded samples without lossy coding.
- Lossless compression (for example FLAC) restores the same PCM values.
- Lossy codecs such as MP3 discard information to reduce storage/bandwidth.
- Use access derivatives without overwriting a managed source/master.

Different copies can serve preservation and online access.

WAV and FLAC

- WAV is a container commonly used for uncompressed PCM audio.
- Its extension alone does not prove that every WAV uses PCM.
- FLAC compresses losslessly and can reconstruct decoded PCM exactly.
- Follow documented institutional format and preservation requirements.

Check file properties—not just the extension.

MP3, AAC and Ogg

- MP3 and AAC are often used for smaller online listening copies.
- Lossy compression can create audible artefacts at low bitrates.
- Ogg is a container that may contain Vorbis or Opus audio codecs.
- Evaluate compatibility, bandwidth and listening quality for your audience.

A small access file is not a replacement for the full source.

Plan your recording before pressing Record

- Identify the audience, purpose, script or questions and required permission.
- Choose a quiet, suitable space and microphone position.
- Check input device, power, storage and recording settings.
- Capture ten seconds of test speech, then listen and adjust.

Preparation prevents avoidable recording failures.

Recording gain and clipping

- Gain controls input amplification; levels should be high enough for useful speech.
- If peaks exceed the available digital range, clipping causes distortion.
- Turning clipped audio down later does not recreate the missing peaks.
- Monitor with headphones and aim for sensible headroom.

Capture clean sound at the source.

Basic editing actions

- Trim only material that is genuinely unwanted for the purpose.
- Split clips, remove accidental long gaps and create gentle fades if needed.
- Apply gain or normalisation while checking for clipping.
- Preserve a full, controlled copy when recording context is important.

Edit a working copy and describe what changed.

Normalisation versus noise reduction

- Peak normalisation adjusts overall gain to a chosen peak level.
- Loudness normalisation targets perceived loudness; it is not identical.
- Noise reduction can reduce steady hiss/hum using a noise-only reference.
- Strong filtering can create metallic artefacts or erase weak speech.

Cleaner audio should not mean altered or fabricated content.

A first edit in Audacity

- Import your sample WAV or record a short authorised message.
- Listen to the full track; identify silence, hiss, clipping and speech sections.
- Trim or fade a working copy, apply only modest justified processing.
- Export a separate access copy and listen to the finished file.

An export is complete only after quality checking.

Master, working file and access copy

- Master/source: retain a controlled, complete recording with documentation.
- Working file: perform edits without overwriting the master.
- Access file: smaller MP3/AAC or another suitable listening derivative.
- Use stable identifiers and record the relationships between versions.

Version clarity protects users and records.

What is a transcript?

- A transcript represents spoken words and relevant non-speech information in text.
- Transcripts support search, quotation, translation and silent access.
- Identify speakers where appropriate and label uncertain words honestly.
- Verify important names, dates and numbers against the actual sound.

Transcription must be checked, not simply generated.

ASR and multilingual challenges

- Automatic speech recognition may misunderstand dialect, names and code-switching.
- Kiswahili and English can occur in the same conversation in Tanzania.
- Mark uncertain passages instead of inventing a plausible phrase.
- Keep a link between the original language, corrected transcript and translation.

Machine-produced text is a draft, not verified evidence.

Audio accessibility

- Provide a usable transcript for prerecorded spoken audio-only resources.
- Offer labelled playback controls; avoid forcing autoplay on users.
- Use clear titles, language labels, duration and navigable section headings.
- For video with meaningful sound, synchronised captions are a separate need.

Accessibility should be designed into the resource.

Metadata for sound recordings

- Record item ID, title, date, source/creator, language and duration.
- Record codec, sample rate, bit depth, channels and version role where needed.
- Describe speakers, context, transcript, permission and access conditions.
- Link compressed derivatives back to the controlled source/master.

Metadata makes audio intelligible, findable and manageable.

Consent, privacy and copyright

- Permission to record is not automatically permission to publish.
- Review personal data, sensitive interviews, music and included quotations.
- Document scope and conditions of permission; apply access restrictions.
- Avoid uploading restricted files to unapproved online editors or ASR tools.

Treat human voices as contextual and potentially sensitive records.

Digital audio preservation

- Keep managed masters with clear provenance and stable identifiers.
- Use backups, access controls, format documentation and preservation planning.
- Checksums identify byte changes; they do not establish truthful origin by themselves.
- Historical tapes may require expert transfer methods and handling.

Long-term access needs more than one saved copy.

Guided practical: audio resource package

- Use the synthetic sample or an authorised 30–60 second recording.
- Prepare an access file and check it on a second playback device if possible.
- Submit the checked transcript, edit log and descriptive metadata.
- Provide a BLISS guide or BRIM archive scenario rationale.

Your output is the whole resource package, not only a sound file.

Applied decision: library guide

- A new student wants directions, library services and accessible help.
- Use concise narration, optional MP3 delivery and an accurate transcript.
- Label duration and language; avoid autoplay and unnecessary bandwidth.
- Test whether the listener can locate the intended information.

Good library audio supports a specific user task.

Applied decision: oral-history exhibit

- A fictional archive has a consented interview with contextual pauses.
- Preserve a full controlled source and document any shortened excerpt.
- Attach provenance, transcript status, rights and access restrictions.
- Do not publish unapproved details or portray an edit as the full recording.

Context, consent and integrity shape archival access.

Unit recap and self-check

- Explain how sound is sampled, encoded, edited and delivered.
- Calculate PCM storage; compare uncompressed, lossless and lossy files.
- Distinguish transcript verification from automated recognition.
- Describe one professional audio workflow for BLISS and one for BRIM.

Record faithfully, publish accessibly, preserve responsibly.